PREMARKET NOTIFICATION
510(k) Submission

GlucaWave™ Wearable Continuous Glucose Monitor

Model GW-2000 Series

Submitter:	NovaSense Biomedical, Inc.
Submission Type:	Traditional 510(k)
Device Classification:	Class II — Product Code: MQF
Regulation:	21 CFR § 880.2860
Predicate Device:	Dexcom G6 CGM System (K182206)
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Prepared By:	Regulatory Affairs Division
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Confidentiality:	Trade Secret / Confidential Commercial Information

TRADE SECRET / CONFIDENTIAL
This document contains proprietary information exempt from disclosure under 21 CFR § 20.61.

1. Executive Summary

1.1 Device Description

NovaSense Biomedical, Inc. submits this Premarket Notification pursuant to Section 510(k) of the Federal Food, Drug, and Cosmetic Act (FD&C Act) and 21 CFR Part 807, Subpart E, for the GlucaWave™ Wearable Continuous Glucose Monitor, Model GW-2000 Series (hereinafter "GlucaWave" or "the Device"). The GlucaWave is a prescription-use, wearable electrochemical biosensor system designed for the continuous, real-time measurement of interstitial fluid (ISF) glucose concentration in adult patients (≥18 years) with diabetes mellitus.

The GlucaWave system consists of three integrated subsystems: (1) a flexible, adhesive-mounted transcutaneous amperometric biosensor filament that is inserted approximately 5–7 mm into the subcutaneous tissue of the abdomen or upper arm; (2) a reusable, Bluetooth Low Energy (BLE 5.2)-enabled transmitter pod housing the signal acquisition, analog-to-digital conversion, and wireless communication electronics; and (3) a companion mobile application (iOS 16+ / Android 13+) that receives, displays, stores, and analyzes glucose data. The system provides continuous glucose readings at 5-minute intervals, with a functional wear duration of 14 days per sensor.

The system incorporates a third-generation electrochemical glucose oxidase (GOx) biosensor fabricated on a platinum-iridium (90/10) working electrode with an electrodeposited Nafion® interference-rejection outer membrane. Signal processing employs a proprietary adaptive two-point factory calibration algorithm with dynamic temperature compensation, eliminating the requirement for fingerstick calibration during the 14-day wear period.

1.2 Intended Use

The GlucaWave™ Wearable Continuous Glucose Monitor is indicated for use by adults (≥18 years) with diabetes mellitus as a replacement for fingerstick blood glucose testing for diabetes management decisions, including insulin dosing. The device is intended to be used with compatible insulin delivery systems or as a standalone monitoring device. The GlucaWave is a prescription device intended for use in the home and clinical environments. The device is not intended for use in critically ill patients, for neonatal glucose monitoring, or for use during magnetic resonance imaging (MRI) procedures.

Regulatory Classification: Product Code MQF; 21 CFR § 880.2860 (glucose test system); Class II; Special Controls per 21 CFR § 880.2860(b) and the FDA Guidance Document "Continuous Glucose Monitors — Submission Guidance" (January 2021).

1.3 Predicate Device and Basis for Substantial Equivalence

The GlucaWave is compared to the predicate device, the Dexcom G6 Continuous Glucose Monitoring System (510(k) K182206; cleared February 2018), a legally marketed Class II continuous glucose monitor indicated for adult and pediatric patients with diabetes. The GlucaWave shares the same intended use, technological characteristics (amperometric electrochemical biosensor, subcutaneous ISF measurement, wireless data transmission), and safety profile as the predicate.

NovaSense asserts that the GlucaWave is substantially equivalent to the predicate based on: (a) identical intended use for adult diabetes management and glycemic decision support; (b) same type of technology — transcutaneous electrochemical glucose biosensor with enzymatic reaction; and (c) comparable or superior performance characteristics as demonstrated by bench testing, clinical performance studies, and validation data presented herein. Differences from the predicate (14-day wear vs. 10-day; no required calibration; BLE 5.2 vs. BLE 4.2; adaptive temperature compensation) do not raise new safety or effectiveness questions, as these represent incremental technological refinements consistent with the state of the art and are supported by performance data.

2. Device Description

2.1 Technical Specifications

Parameter	Specification
Sensor Type	Amperometric electrochemical — glucose oxidase (GOx) enzymatic
Working Electrode	Platinum-iridium (Pt90/Ir10) wire, 0.25 mm diameter, tip area 0.008 cm²
Counter/Reference Electrode	Ag/AgCl printed electrode on flexible polyimide substrate
Outer Membrane	Electrodeposited Nafion® (sulfonated PTFE), 8 µm nominal thickness
Glucose Measurement Range	40–400 mg/dL (2.2–22.2 mmol/L)
Measurement Interval	Every 5 minutes (288 readings/day)
Sensor Wear Duration	14 days (336 hours), single use per sensor
Insertion Depth	5–7 mm subcutaneous (adjustable per body habitus)
Transmitter Dimensions	38.5 mm × 28.0 mm × 12.5 mm; mass 9.8 g
Operating Temperature	10°C – 40°C (sensor); 0°C – 45°C (transmitter)
Transmitter Battery	Lithium-ion, 180 mAh; recharges via USB-C, 14-day continuous use
Wireless Protocol	Bluetooth Low Energy (BLE) 5.2, 2.4 GHz ISM band, AES-128 encryption
Display Range	Software: 40–400 mg/dL; trend arrows at 1–3 mg/dL/min rate of change
Ingress Protection	Transmitter: IP28 (water-resistant to 2.4 m for up to 24 hours)
EMC Standards	IEC 60601-1-2:2014+AMD1:2020; FCC Part 15B; IC RSS-247
Electrical Safety	IEC 60601-1:2005+AMD1:2012+AMD2:2020; applied part Type BF

2.2 Biosensor Mechanism

The GlucaWave employs a three-electrode amperometric biosensor configuration operating on the electrochemical oxidation of hydrogen peroxide (H_2O_2) generated by the enzymatic reaction between glucose and dissolved oxygen, catalyzed by glucose oxidase (GOx) immobilized within a crosslinked glutaraldehyde/bovine serum albumin (GA/BSA) matrix on the working electrode surface. The electrochemical cascade proceeds as follows:

$$\beta\text{-D-Glucose} + O_2 \xrightarrow{(GOx)} \text{D-glucono-}\delta\text{-lactone} + H_2O_2 \quad \text{[Reaction 1]}$$

$$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^- \quad \text{[Reaction 2, at +0.60V vs. Ag/AgCl]}$$

The resultant anodic current is linearly proportional to glucose concentration across the measurement range ($r^2 \geq 0.998$ over 40–400 mg/dL). The multilayer electrode architecture (innermost to outermost) comprises: (1) Pt/Ir wire working electrode; (2) electropolymerized poly(phenylenediamine) (PPD) exclusion layer (80–120 nm) blocking electroactive interferents (ascorbic acid, uric acid, acetaminophen); (3) GOx enzyme matrix (~4 µm); (4) polyurethane glucose-limiting outer membrane (12 µm) providing linear response control and oxygen balance; and (5) Nafion® outer layer (8 µm) providing additional interferent exclusion and biocompatibility.

A temperature-sensing thermistor embedded in the transmitter pod provides continuous temperature data (±0.2°C accuracy) used by the signal processing algorithm to compensate for temperature-dependent enzyme kinetics (Km shift ~0.8 mM/°C) and membrane permeability changes across the operating range.

2.3 Signal Processing Architecture

Raw amperometric current (nA range) is acquired by the transmitter's application-specific integrated circuit (ASIC) at 1 Hz and subjected to a multistage signal conditioning pipeline:

- Stage 1 — Analog Front-End: Transimpedance amplifier (TIA) converts working electrode current to voltage (gain: 10 MΩ); 16-bit ADC digitization at 1 Hz sampling.
- Stage 2 — Digital Filtering: 4th-order low-pass Butterworth filter (cutoff: 0.02 Hz) attenuates motion artifact and high-frequency noise; Kalman filter for state estimation.
- Stage 3 — Factory Calibration Application: A two-point factory calibration model (established during manufacturing at 100 mg/dL and 250 mg/dL reference standards under controlled ISF-mimic conditions) maps raw current to interstitial glucose values using sensor-specific sensitivity (slope) and background current (intercept) coefficients stored in encrypted non-volatile memory.
- Stage 4 — Temperature Compensation: Adaptive correction using real-time thermistor data and a pre-validated polynomial compensation model (third-order, validated 10–40°C).
- Stage 5 — Physiological Plausibility Check: Rate-of-change limiting algorithm rejects physiologically implausible glucose excursions (>3.5 mg/dL/min sustained over 5 minutes) as signal artifacts.
- Stage 6 — BLE Transmission: AES-128 encrypted, authenticated 5-minute glucose readings transmitted to the companion application; readings retained in transmitter FIFO buffer for up to 3 hours if BLE connection is interrupted.

3. Substantial Equivalence Analysis

The following comparative analysis demonstrates that the GlucaWave™ GW-2000 is substantially equivalent to the predicate device (Dexcom G6, K182206) pursuant to 21 CFR § 807.87(f) and FD&C Act § 513(i). Both devices share the same intended use for continuous glucose monitoring in adults with diabetes. Technological differences are identified and demonstrated not to raise new questions of safety or effectiveness.

Technical Parameter	GlucaWave GW-2000 (Subject Device)	Dexcom G6 (Predicate — K182206)	Equivalence Determination
Intended Use	Adults ≥18 yrs; diabetes management; insulin dosing replacement	Adults ≥18 yrs; diabetes management; insulin dosing replacement	**EQUIVALENT**
Sensing Technology	Electrochemical amperometric; GOx enzyme; Pt/Ir working electrode; three-electrode cell	Electrochemical amperometric; GOx enzyme; Pt working electrode; three-electrode cell	**EQUIVALENT**
Measurement Site & Depth	Abdomen or upper arm; 5–7 mm subcutaneous (ISF)	Abdomen or back of upper arm; 5–7 mm subcutaneous (ISF)	**EQUIVALENT**
Wear Duration	14 days per sensor (336 hr)	10 days per sensor (240 hr)	**DIFFERENT — See Note A**
Calibration Requirement	Factory calibrated; no fingerstick required	Factory calibrated; no fingerstick required	**EQUIVALENT**
Wireless Communication	BLE 5.2; AES-128 encryption; 10 m nominal range	BLE 4.2; AES-128 encryption; 6 m nominal range	**DIFFERENT — See Note B**
Measurement Range & Interval	40–400 mg/dL; 5-min intervals	40–400 mg/dL; 5-min intervals	**EQUIVALENT**
Biocompatibility	ISO 10993 compliant; skin-contact materials: medical-grade silicone, acrylate adhesive, PA 6.6 filament sheath	ISO 10993 compliant; skin-contact materials: silicone, acrylate adhesive	**EQUIVALENT**

Note A — 14-Day Wear Duration

The extended 14-day sensor wear duration (vs. 10-day predicate) represents an incremental improvement enabled by the addition of a crosslinked polyurethane glucose-limiting membrane with enhanced hydrolytic stability (validated per ASTM F1980 accelerated aging to 24-month equivalent). NovaSense conducted a 160-subject clinical performance study (NovaSense Protocol NS-CGM-003) demonstrating non-inferior MARD performance at Days 11–14 (MARD = 9.1%) compared to Days 1–10 (MARD = 8.6%), with no increase in adverse event rate attributable to extended wear. This difference does not raise new safety or effectiveness concerns beyond those applicable to the predicate.

Note B — BLE 5.2 Communication Protocol

The upgrade from BLE 4.2 (predicate) to BLE 5.2 enhances communication range, throughput, and connection robustness. Security specifications are equivalent (AES-128 encryption; mutual authentication). The BLE 5.2 implementation was validated for electromagnetic compatibility per IEC 60601-1-2:2014+AMD1:2020 and FCC Part 15B. Cybersecurity controls have been implemented per FDA's "Cybersecurity in Medical Devices" Final Guidance (September 2023), including secure boot, cryptographically signed firmware updates, and anomaly detection. This technological difference does not introduce new risks.

4. Biocompatibility Assessment

Biocompatibility evaluation was conducted in accordance with ISO 10993-1:2018 "Biological Evaluation of Medical Devices — Part 1: Evaluation and Testing Within a Risk Management Process" and the FDA Guidance "Use of International Standard ISO 10993-1, Biological Evaluation of Medical Devices — Part 1" (September 2020). The GlucaWave sensor and transmitter contact materials were categorized by nature of body contact and duration:

- Sensor filament assembly: Prolonged surface contact / tissue-implanted contact (>24 hours; transcutaneous); classified as Prolonged Contact, Implant.
- Adhesive patch (acrylate-based): Prolonged surface contact with intact skin (>24 hours; ≤30 days).
- Transmitter housing (polycarbonate/ABS over-mold): External communicating device; not in direct patient contact; no biocompatibility testing required per ISO 10993-1 Table A.1.

ISO 10993 Test	Standard	Result	Disposition
Cytotoxicity (MEM Elution)	ISO 10993-5	Grade 0 (Non-cytotoxic); cell viability ≥95% vs. negative control	**PASS**
Sensitization (GPMT)	ISO 10993-10	Induction/challenge score = 0; non-sensitizer classification	**PASS**
Intracutaneous Reactivity	ISO 10993-10	Mean erythema/edema scores: 0.0 (test) vs. 0.0 (blank); no significant difference	**PASS**
Acute Systemic Toxicity (IV)	ISO 10993-11	No systemic adverse effects; all animals survived; body weight within 5% of controls	**PASS**
Subacute Toxicity (28-day implant, rat)	ISO 10993-11	No material-related histopathological findings; mild transient foreign body reaction consistent with device trauma	**PASS**
Genotoxicity (Ames + in vitro MN)	ISO 10993-3	Non-mutagenic (Ames); non-clastogenic (micronucleus, L5178Y); both negative	**PASS**
Hemocompatibility	ISO 10993-4	Hemolysis index = 1.2% (<5% criterion); no thrombogenic activity (APTT/PT within ±10% of control)	**PASS**
Implantation (Subcutaneous, 12-week rabbit)	ISO 10993-6	Microscopic score: 2.6 (test) vs. 2.1 (control); difference <1.0 — no significant adverse tissue reaction	**PASS**
Pyrogenicity (Rabbit Pyrogen)	ISO 10993-11	Maximum individual temperature rise <0.6°C; total rise	**PASS**

		<1.4°C; meets USP <151> criteria	
Chemical Characterization (AET/TTC)	ISO 10993-18	All extractables <Analytical Evaluation Threshold (AET); no Threshold of Toxicological Concern (TTC) exceedances identified	**PASS**

All testing was performed by accredited, ISO 17025-compliant contract research organizations (Pacific BioLabs; Nelson Laboratories). Full biocompatibility test reports are provided in Appendix D. The biocompatibility risk assessment, performed in accordance with ISO 10993-17 for toxicological risk assessment, concludes that the GlucaWave™ presents no unreasonable biological risk for its intended prolonged-contact, transcutaneous application.

5. Software Documentation Summary

Software documentation for the GlucaWave™ system has been prepared in accordance with IEC 62304:2006+AMD1:2015 "Medical Device Software — Software Life Cycle Processes" and the FDA Guidance "Software as a Medical Device (SaMD): Clinical Evaluation" (2017), and the FDA Guidance "Deciding When to Submit a 510(k) for a Change to an Existing Device" (October 2017). A Software Bill of Materials (SBOM) has been provided in accordance with FDA's "Cybersecurity in Medical Devices" Final Guidance (September 2023).

5.1 Software Safety Classification

Software safety classification was determined per IEC 62304 § 4.3 based on the severity of injury that could result from software failure:

- Transmitter Firmware (GW-FW-2000): CLASS C — Incorrect glucose readings resulting from firmware failure could cause the user to make an incorrect insulin dosing decision, resulting in severe hypoglycemia or hyperglycemia, which could be life-threatening. Software failure can result in SERIOUS INJURY or DEATH.
- Companion Mobile Application (GlucaWave App v3.x): CLASS C — The application is used to display glucose values and trend arrows that directly inform insulin dosing decisions. Failure to display or incorrect display of glucose data carries the same risk profile as transmitter firmware.
- Manufacturing/Calibration Software (GW-CAL-1000): CLASS B — Software failure during factory calibration could result in systematically biased sensors, potentially requiring batch recall. Injury from a single device is possible but not life-threatening in isolation.

5.2 Software Development Lifecycle

The GlucaWave software was developed under a formal IEC 62304-compliant Software Development Lifecycle (SDLC) including:

- Software Development Plan (SDP-GW-001 Rev C): Defines development methodology (Agile-SAFE adapted for medical device), version control policy (Git with signed commits; GitLab CI/CD), and risk-based testing approach.
- Software Requirements Specification (SRS-GW-001 Rev D): 847 functional and non-functional requirements; 100% requirements traceability to design, test, and risk management artifacts via electronic SDLC platform (MedAccred QMS).
- Software Architecture Document (SAD-GW-001 Rev B): Documents firmware microarchitecture (RTOS: FreeRTOS v10.5.1 on ARM Cortex-M33); application software stack (React Native 0.72 on iOS/Android); separation of safety-critical and non-safety-critical software units.
- Verification & Validation (V&V Report SVR-GW-001 Rev A): 2,341 test cases executed; 100% pass rate; 0 open critical or major defects at release. Regression test suite executed on each firmware/application release candidate.
- Software Maintenance Plan: Defines post-market surveillance triggers for software updates; all Class C software changes require 510(k) change evaluation per FDA guidance.

5.3 Cybersecurity Controls

Per FDA's Final Cybersecurity Guidance (2023), the following security controls have been implemented and verified: secure boot with RSA-2048 signature verification; AES-128 BLE session encryption with ECDH key exchange; over-the-air (OTA) firmware updates restricted to cryptographically signed packages from NovaSense-controlled update servers; network penetration testing conducted by independent third-party (Rapid7; Report CYB-GW-2025-003); CVE monitoring and patch policy (critical patches within 30 days); and SBOM provided in SPDX 2.3 format (Appendix G).

6. Performance Testing

Analytical and clinical performance was evaluated through a combination of bench/laboratory testing and a pivotal clinical performance study (Protocol NS-CGM-003; 18 U.S. investigational sites; NCT Registration: NCT05887234) enrolling 160 adult subjects with Type 1 and Type 2 diabetes mellitus. All clinical testing was conducted under IDE exemption per 21 CFR § 812.2(c)(3). Performance acceptance criteria were established a priori and benchmarked against the FDA CGM Guidance performance benchmarks (January 2021).

6.1 Accuracy (MARD — Clinical Study)

Accuracy Metric	GlucaWave Result	Acceptance Criterion	Status
Overall MARD (all glucose ranges)	8.8%	≤10.8% (FDA Benchmark)	**PASS**
MARD — Hypoglycemic Range (<70 mg/dL)	9.6%	≤15%	**PASS**
MARD — Euglycemic Range (70–180 mg/dL)	8.4%	≤12%	**PASS**
MARD — Hyperglycemic Range (>180 mg/dL)	9.2%	≤12%	**PASS**
% Readings within ±15 mg/dL or ±15% (lower of)	94.3%	≥87%	**PASS**
% Readings within ±20 mg/dL or ±20% (lower of)	98.7%	≥95%	**PASS**
Clarke Error Grid Zone A+B	99.8%	≥99%	**PASS**
Lag Time (ISF vs. capillary blood)	3.9 min (median)	≤10 min	**PASS**
MARD — Day 11–14 (Extended Wear)	9.1%	≤12% (internal criterion)	**PASS**

6.2 Precision (Bench Testing — Dual-Sensor Simultaneous Wear)

Precision was assessed by simultaneous dual-sensor wear in 40 subjects (20 T1D, 20 T2D) over 14 days, with paired glucose readings compared at matched time points. Additionally, in vitro precision testing was performed using a calibrated peristaltic pump-based ISF analog (PBS/glucose, pH 7.4, 37°C ± 0.5°C):

- Within-run CV (in vitro, n=20 replicates at each concentration): 100 mg/dL: CV = 1.8%; 200 mg/dL: CV = 1.6%; 350 mg/dL: CV = 2.1%. Acceptance criterion: CV ≤5.0%. All concentrations PASS.
- Between-sensor CV (clinical, n=40 paired sensors, 14-day wear): Overall CV = 4.2% (acceptance criterion: ≤8.0%). PASS.
- Lot-to-lot variability (3 manufacturing lots, n=30 sensors/lot): Inter-lot MARD difference ≤0.6% across all lots (acceptance criterion: ≤2.0%). PASS.

6.3 Interference Testing

Endogenous and exogenous interferent testing was conducted per CLSI EP7-A3 "Interference Testing in Clinical Chemistry" and the FDA CGM Guidance. Interferent solutions were prepared in ISF-analog and tested at clinically relevant concentrations using an in vitro flow-cell biosensor test system:

Interferent	Test Concentration	Bias Observed	Status
Ascorbic Acid	0.3 mM (physiologic max)	+2.1 mg/dL (0.9%)	**PASS**

Acetaminophen	0.33 mM (therapeutic max)	+3.4 mg/dL (1.5%)	**PASS**
Uric Acid	0.6 mM (hyperuricemic max)	+1.8 mg/dL (0.8%)	**PASS**
Dopamine	4 µM	+0.7 mg/dL (<1%)	**PASS**
Ibuprofen	0.24 mM (therapeutic max)	-1.2 mg/dL (0.5%)	**PASS**
Hydroxyurea	0.5 mM	+4.1 mg/dL (1.8%)	**PASS**
Maltose (IV nutrition)	0.5 mM	+1.3 mg/dL (0.6%)	**PASS**
Hematocrit (20–60%)	Full range tested	Max bias: +2.9% at Hct = 20%	**PASS**

Acceptance criterion for all interferents: bias ≤±10 mg/dL at glucose ≤100 mg/dL or ±10% at glucose >100 mg/dL. All interferents tested at or above maximum physiological/pharmacological concentrations. All results within acceptance criteria.

7. Labeling Requirements — 21 CFR Part 801 Compliance

Device labeling has been developed in compliance with 21 CFR Part 801 (Labeling), 21 CFR § 801.109 (Prescription devices), and the FDA CGM Labeling Guidance. Labeling components include: (1) Device Label (sensor package; transmitter; charging case); (2) Instructions for Use (IFU; DFU); (3) Companion Application User Interface Text; (4) Physician Prescribing Information (PPI); and (5) Promotional Labeling. All labeling has been reviewed and approved by Regulatory Affairs and subjected to human factors validation (see Human Factors Summary, Appendix E).

Labeling Requirement	21 CFR / Guidance Citation	Status
Device name (proprietary and common/usual name)	21 CFR § 801.61	**COMPLIANT**
Manufacturer name and principal place of business	21 CFR § 801.1	**COMPLIANT**
Net quantity of contents	21 CFR § 801.62	**COMPLIANT**
Adequate directions for use	21 CFR § 801.5	**COMPLIANT — Rx exemption per § 801.109**
Prescription ('Rx Only') designation	21 CFR § 801.109(b)(1)	**COMPLIANT**
Intended use and indications for use statement	21 CFR § 801.4; CGM Guidance §V.A	**COMPLIANT**
Contraindications (MRI, pediatric, critically ill)	21 CFR § 801.57; CGM Guidance §V.B	**COMPLIANT**
Warnings and Precautions (hypoglycemia symptoms, lag time, interference)	21 CFR § 801.57	**COMPLIANT**
Accuracy performance summary (MARD, Clarke EGA)	FDA CGM Guidance §V.D	**COMPLIANT**
Calibration instructions (factory cal; no fingerstick required)	FDA CGM Guidance §V.A.4	**COMPLIANT**
Device wear duration limitation (14 days)	FDA CGM Guidance §V.A.3	**COMPLIANT**
Water resistance specifications (IP28 rating; max 2.4 m / 24 hr)	IFU Section 6.3	**COMPLIANT**
Electromagnetic interference warnings (MRI; high-field equipment)	IEC 60601-1-2; IFU Section 7.2	**COMPLIANT**
Storage and handling conditions (temperature, humidity, shelf life)	21 CFR § 801.1; IFU Section 2.1	**COMPLIANT**
UDI — Device Identifier (DI) on label and IFU	21 CFR Part 830; 21 CFR § 801.20	**COMPLIANT — GS1 GTIN assigned**
UDI-PI (lot/batch number, serial number, mfg/exp dates)	21 CFR § 801.40–801.50	**COMPLIANT**
Symbol glossary (ISO 15223-1 symbols with legend)	FDA Symbols Guidance (2019); ISO 15223-1:2021	**COMPLIANT**
Human factors usability validation of IFU	FDA HF Guidance (2016); IEC 62366-1:2015	**COMPLIANT — Summative study completed**

8. Risk Management Summary

Risk management was conducted in full compliance with ISO 14971:2019 "Medical Devices — Application of Risk Management to Medical Devices" and the FDA Guidance "Factors to Consider When Making Benefit-Risk Determinations in Medical Device Premarket Approval and De Novo Classifications" (August 2019). A comprehensive Risk Management File (RMF-GW-001 Rev C) has been established encompassing hazard identification, risk estimation, risk evaluation, risk control, and residual risk assessment for the complete GlucaWave system including hardware, software, and use-related risks.

Severity and Probability scales are defined as follows: Severity — S1: Negligible (no injury); S2: Minor (reversible, no medical intervention); S3: Serious (hospitalization or prolonged medical intervention); S4: Critical (permanent impairment); S5: Catastrophic (death). Probability — P1: Improbable ($<1\times10^{-6}$/device-year); P2: Remote (1×10^{-6} to 1×10^{-4}); P3: Occasional (1×10^{-4} to 1×10^{-2}); P4: Probable ($>1\times10^{-2}$). Risk acceptability matrix follows ISO 14971 Annex C; risks with Residual Risk Index (RRI = S × P class) of ≥S3/P3 or higher require additional verification of risk control effectiveness.

Hazard / Harm	Failure Mode	Pre-Control Risk (S/P)	Risk Controls Implemented	Residual Risk (S/P)	Accept-able?
Severe Hypoglycemia / Death	False HIGH glucose reading → missed or delayed low glucose alert → patient fails to treat hypoglycemia	**S5/P3**	• Multi-layer electrochemical interferent exclusion (PPD + Nafion layers) • Physiological plausibility checking algorithm (Stage 5 signal processing) • Low Glucose Alert (LGA) and Urgent Low Glucose Alert (55 mg/dL threshold, non-silenceable) • IFU warning to verify with fingerstick if symptomatic • Clinical MARD ≤10% validation across hypoglycemic range	**S5/P2**	**YES**
Severe Hyperglycemia / DKA	False LOW glucose reading → undetected hyperglycemia → insufficient insulin → diabetic ketoacidosis	**S4/P3**	• Factory two-point calibration with ±5% accuracy acceptance at 400 mg/dL • 14-day wear limit enforced by sensor expiry interrupt in transmitter firmware • High Glucose Alerts configurable by prescriber (default: 250 mg/dL) • IFU guidance on symptoms of hyperglycemia requiring clinical evaluation	**S4/P2**	**YES**
Skin Injury / Allergic Contact Dermatitis	Allergic or irritant reaction to adhesive patch or sensor filament sheath materials	**S3/P3**	• ISO 10993 biocompatibility qualification (sensitization GPMT: Pass; intracutaneous reactivity: Pass) • ISO 10993-18 chemical characterization — no sensitizing extractables above AET • IFU contraindication for known acrylate/silicone hypersensitivity • Post-market surveillance: adverse event monitoring per 21 CFR Part 803	**S2/P3**	**YES**
Infection at Insertion Site	Introduction of pathogen during transcutaneous sensor insertion	**S4/P2**	• Sensor filament and inserter needle sterile-packaged (EO sterilization per ISO 11135; SAL $\leq10^{-6}$)	**S3/P1**	**YES**

	→ local or systemic infection		• Single-use, sterile, retractable automated inserter eliminates user handling of needle • PA 6.6 filament sheath antimicrobial surface treatment (validated per ISO 22196) • IFU: site preparation with isopropyl alcohol; contraindication for insertion over compromised skin		
Insulin Dosing Error Due to Software Failure	Transmitter firmware crash or companion app display error results in missing or incorrect glucose value used for insulin bolus calculation	**S5/P3**	• IEC 62304 Class C software qualification; >2,300 test cases; 0 open critical defects • Watchdog timer with automatic firmware recovery (validated MTTR <30 seconds) • On-screen warning: 'No Readings — Check Sensor' displayed within 6 minutes of data loss • FIFO buffer retains up to 3 hours of readings for retrospective display on reconnection • User training module: verify with fingerstick when CGM unavailable before dosing	**S5/P2**	**YES**

Overall Residual Risk Conclusion: Following implementation of all risk controls identified in the Risk Management File (RMF-GW-001 Rev C), the overall residual risk of the GlucaWave™ GW-2000 system has been evaluated and determined to be ACCEPTABLE relative to the established benefit-risk profile. The medical benefits of continuous, real-time glucose monitoring for diabetes management — including reduction in HbA1c, reduction in hypoglycemic events, and improved time-in-range — substantially outweigh the residual risks after control measures. This conclusion is consistent with the benefit-risk framework of ISO 14971:2019 § 9 and is supported by clinical evidence from study NS-CGM-003 and published literature. No new residual risks were identified that would preclude clearance of this device.

END OF REGULATORY SUBMISSION DOCUMENTATION GUIDE

GlucaWave™ GW-2000 — 510(k) Premarket Notification — NovaSense Biomedical, Inc. — March 2026

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